SUMMARY

(1) A **binary operation** $*$ on a non-empty set S is a rule, which associates to each ordered pair (a,b) of elements a,b in S an unique element $a*b$ in S.

(2) **Commutative property:** Any binary operation $*$ defined on a nonempty set S is said to satisfy the commutative property, if $a*b=b*a,\ \forall a,b\in S$.

(3) **Associative property:** Any binary operation $*$ defined on a nonempty set S is said to satisfy the associative property, if $a*(b*c)=(a*b)*c,\ \forall a,b,c\in S$.

(4) **Existence of identity property:** An element $e\in S$ is said to be the **Identity Element** of S under the binary operation $*$ if for all $a\in S$ we have that $a*e=a$ and $e*a=a$.

(5) **Existence of inverse property:** If an identity element e exists and if for every $a\in S$, there exists b in S such that $a*b=e$ and $b*a=e$ then $b\in S$ said to be the **Inverse Element** of a. In such instance, we write $b=a^{-1}$.

(6) **Uniqueness of Identity:** In an algebraic structure the identity element (if exists) must be unique.

(7) **Uniqueness of Inverse:** In an algebraic structure the inverse of an element (if exists) must be unique.

(8) A **Boolean Matrix** is a real matrix whose entries are either 0 or 1.

(9) **Modular arithmetic:** Let n be a positive integer greater than 1 called the **modulus**. We say that two integers a and b are congruent modulo n if $b-a$ is divisible by n. In other words $a\equiv b\,(\text{mod n})$ means $a-b=n\cdot k$ for some integer k and b is the **least non-negative integer reminder** when a is divided by n. $(0\le b\le n-1)$

(10) Mathematical logic is a study of reasoning through mathematical symbols.

(11) Let p be a simple statement. Then the **negation** of p is a statement whose truth value is opposite to that of p. It is denoted by $\neg p$, read as **not** p. The truth value of $\neg p$ is T, if p is F, otherwise it is F.

(12) Let p and q be any two simple statements. The **conjunction** of p and q is obtained by connecting p and q by the word **and**. It is denoted by $p\wedge q$, read as 'p conjunction q' or 'p hat q'. The truth value of $p\wedge q$ is T, whenever both p and q are T and it is F otherwise.

(13) The **disjunction** of any two simple statements p and q is the compound statement obtained by connecting p and q by the word 'or'. It is denoted by $p\vee q$, read as 'p disjunction q' or 'p cup q'. The truth value of $p\vee q$ is F, whenever both p and q are F and it is T otherwise.

(14) The **conditional statement** of any two statements p and q is the statement, 'If p, then q' and it is denoted by $p\to q$. The statement $p\to q$ has a truth value F when q has the truth value F and p has the truth value T; otherwise it has the truth value T.

(15) The **bi-conditional statement** of any two statements p and q is the statement 'p if and only if q' and is denoted by $p\leftrightarrow q$ The statement $p\leftrightarrow q$ has the truth value T whenever both p and q have identical truth values; otherwise has the truth value F.

(16) A statement is said to be a **tautology** if its truth value is always T irrespective of the truth values of its component statements. It is denoted by $\mathbb{T}$.

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